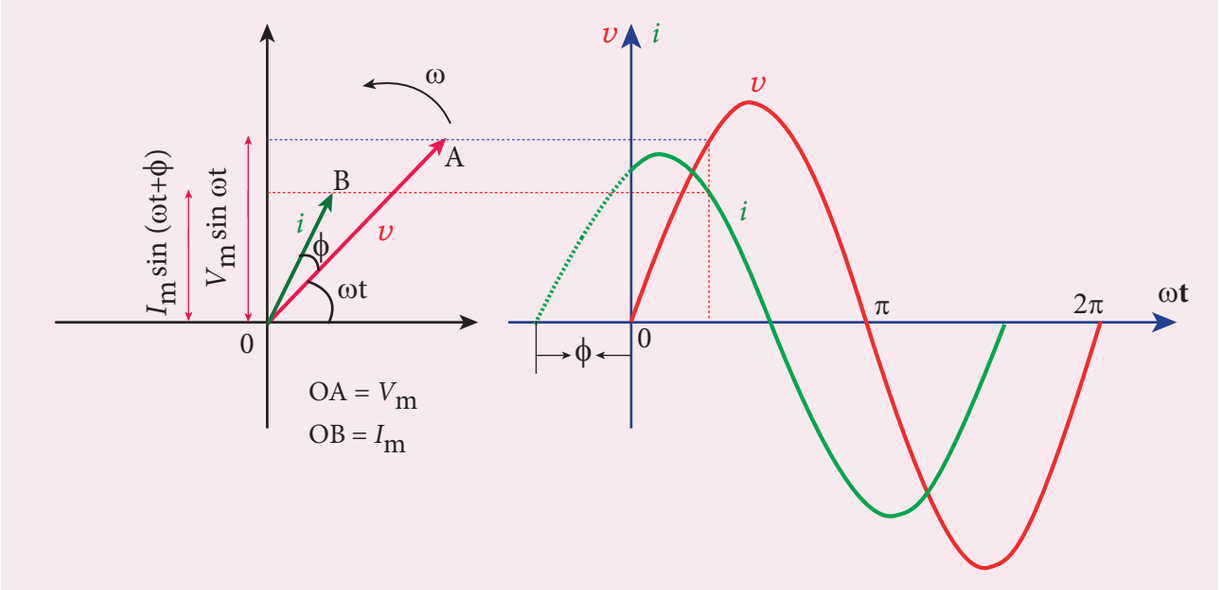


Figure 4.39 Phasor diagram and wave diagram say that i leads υ by ϕ

another phasor $\overrightarrow{OB}$. Here ϕ is the phase angle between voltage and current. In this case, the current leads the voltage by phase angle ϕ which is shown in Figure 4.39. If the current lags behind the voltage, then we write $i = I_m \sin(\omega t - \phi)$.

4.7.4 AC circuit containing pure resistor

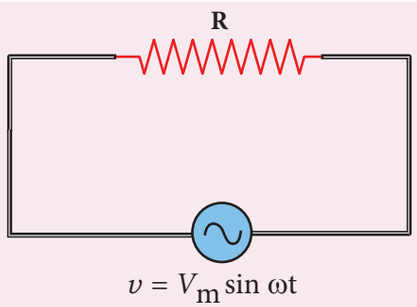


Figure 4.40 AC circuit with resistor

Consider a circuit containing a pure resistor of resistance R connected across an alternating voltage source (Figure 4.40). The instantaneous value of the alternating voltage is given by

$$\upsilon = V_m \sin \omega t \qquad (4.37)$$

An alternating current i flowing in the circuit due to this voltage, develops a potential drop across R and is given by

$$V_R = iR \qquad (4.38)$$

Kirchoff's loop rule (Refer section 2.5.2) states that the algebraic sum of potential differences in a closed circuit is zero. For this resistive circuit,

$$\upsilon - V_R = 0$$

From equation (4.37) and (4.38),

$$V_m \sin \omega t = iR$$

$$i = \frac{V_m}{R} \sin \omega t$$

$$i = I_m \sin \omega t \qquad (4.39)$$

where $\frac{V_m}{R} = I_m$, the peak value of alternating current in the circuit. From equations (4.37) and (4.39), it is clear that

12th Physics_Vol-1_Unit 4_English Medium.indd 238 23-12-2022 17:58:14